

BROOME, Australia

WMO: 942030

Lat: 17.9475S

Lon: 122.2353E

Elev: 9

StdP: 101.22

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	12.1	-7.0	2.1	27.5	-4.1	2.7	26.4	9.3	21.8	8.5	22.7	1.5	110	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.3	37.5	20.9	36.0	21.6	34.9	22.4	28.3	31.8	27.9	31.4	27.6	31.2	4.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.3	23.2	30.6	27.0	22.7	30.5	26.6	22.2	30.2	91.7	31.9	89.7	31.5	88.1	31.3	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	8.0	7.3	DB	7.8	41.2	1.3	1.3	6.9	42.1	6.1	42.8	5.4	43.5	4.5	44.5	(4)
(5)			WB	4.7	29.5	0.9	0.8	4.1	30.0	3.6	30.5	3.1	31.0	2.5	31.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	29.5	29.3	29.5	28.5	24.8	22.3	21.7	22.3	25.0	27.9	29.6	30.1	(6)	
	DBStd	3.69	1.45	1.59	1.63	1.96	2.34	2.37	2.20	2.02	2.16	2.01	1.62	1.53	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	3	0	0	0	0	0	2	1	0	0	0	0	0	(9)	
	CDD10.0	6095	606	539	605	555	459	368	363	382	450	554	589	624	(10)	
	CDD18.3	3056	348	306	347	305	201	119	106	124	200	296	339	366	(11)	
	CDH23.3	36750	4552	3913	4503	3788	2101	1099	1038	1307	1946	3302	4318	4883	(12)	
(13)	CDH26.7	16897	2189	1816	2234	1932	917	358	330	485	758	1359	2043	2477	(13)	
(14)	Wind	WSAvg	3.8	4.5	4.2	3.6	3.1	3.2	3.5	3.3	3.4	3.7	4.2	4.5	4.7	(14)
(15) Precipitation	PrecAvg	660	213	185	105	22	24	15	3	2	1	1	9	67	(15)	
	PrecMax	1481	916	494	493	247	268	208	50	23	24	28	58	408	(16)	
	PrecMin	132	5	6	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	306	223	129	114	40	45	36	7	4	3	4	14	77	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	36.1	37.8	38.1	35.9	33.0	33.0	34.3	37.4	39.0	40.1	38.8	(19)	
		MCWB	25.5	24.2	22.7	21.1	19.1	18.8	17.2	16.4	17.1	19.0	20.2	22.2	(20)	
	2%	DB	34.0	34.1	35.9	36.1	34.1	31.2	31.3	32.5	35.2	36.1	36.1	35.3	(21)	
		MCWB	26.6	26.7	23.8	21.2	18.9	17.7	16.7	16.0	16.8	19.6	21.9	25.0	(22)	
	5%	DB	33.1	33.1	34.4	34.8	32.8	30.1	30.0	31.0	33.0	33.4	34.0	34.0	(23)	
		MCWB	26.8	26.9	25.4	21.5	19.0	17.4	16.4	15.6	17.2	21.9	24.2	25.9	(24)	
	10%	DB	32.4	32.3	33.2	33.6	31.4	28.9	28.5	29.5	30.5	31.8	32.8	33.0	(25)	
		MCWB	26.7	26.7	26.1	22.5	19.3	17.4	16.3	15.7	19.0	23.3	24.9	26.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.6	28.7	29.0	27.7	26.0	24.5	23.2	22.7	24.6	26.6	27.6	28.2	(27)	
		MCDB	32.4	32.0	32.3	31.1	28.8	27.5	25.6	25.8	27.2	29.8	31.9	32.2	(28)	
	2%	WB	28.0	28.3	28.3	27.2	24.9	23.0	21.8	22.0	23.9	26.1	27.0	27.7	(29)	
		MCDB	31.5	31.6	31.7	30.7	28.0	26.3	25.0	25.1	26.6	29.4	31.0	31.6	(30)	
	5%	WB	27.6	27.8	27.8	26.7	24.0	22.0	20.7	21.3	23.4	25.7	26.6	27.2	(31)	
		MCDB	31.2	31.2	31.2	30.4	27.7	25.4	24.6	24.6	26.5	29.1	30.6	31.3	(32)	
	10%	WB	27.2	27.4	27.4	26.2	23.0	20.9	19.7	20.5	22.8	25.3	26.2	26.9	(33)	
		MCDB	30.9	30.9	30.9	30.0	27.0	25.1	24.2	24.2	26.3	28.8	30.2	31.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	5.9	7.7	11.0	13.4	13.6	14.5	15.3	12.2	9.0	7.2	6.3	(35)	
		MCDBR	7.0	6.9	9.8	13.0	15.1	14.7	16.1	17.8	16.7	12.7	10.2	8.2	(36)	
	5% WB	MCWBR	3.2	3.3	4.5	5.9	6.5	6.6	6.9	7.6	7.4	5.6	5.0	3.7	(37)	
		MCWBR	6.5	6.2	7.7	9.0	11.2	11.5	12.7	12.9	10.6	8.2	7.3	6.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.422	0.413	0.390	0.370	0.355	0.331	0.328	0.336	0.367	0.405	0.409	0.409	(40)	
		taud	2.471	2.504	2.566	2.563	2.541	2.582	2.573	2.510	2.468	2.400	2.431	2.493	(41)	
	Ebn at Noon	Ebn at Noon	924	923	922	904	882	888	902	924	929	919	930	936	(42)	
		Edh at Noon	119	114	103	97	93	85	88	101	112	125	123	116	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.56	6.43	6.18	5.92	5.11	4.79	5.08	5.98	6.86	7.41	7.67	7.19	(44)	
	RadStd	RadStd	0.61	0.72	0.66	0.28	0.32	0.31	0.28	0.17	0.17	0.21	0.21	0.43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.62	N/A	N/A	+0.71	+0.39	+0.31	N/A	N/A	+230	+231	(46)
(47)	Regional (3 neighbors)	+0.44	N/A	N/A	+0.54	+0.37	+0.32	N/A	N/A	+230	+231	(47)

Nomenclature: See separate page